Unit 2: Project Life Cycle and Processes

BESE 7th Semester

- **“Every project you have seen, from building a house in your village to developing an app in Pokhara, follows a certain path. What do you think happens between the first idea and the final handover?”**

Why the chapter matters

- **How real projects move from start to end**, and why missing a step leads to delay, cost overruns, and confusion.
- Example:

"In Nepal, several municipal e-governance projects failed because teams jumped into coding before defining the scope. Understanding the life cycle prevents the same mistake in your own work."

Project Life Cycle

- **Example: Building a small mobile app for a local trekking agency in Lakeside, Pokhara.**
- You can show that the project moves through:
  - identifying the need (agency wants online booking)
  - planning (what features, timeline, cost)
  - execution (development and testing)
  - closure (handover, training, and support)

Project Life Cycle

- Project life cycle is a series of phases of a project from initiation to completion.
- The life cycle gives a practical approach to problem solving applied to all aspects of a project.
- Phases in a project life cycle encompasses sequential and overlapping phases.
- A project life cycle typically has 4 major phases:
  - Initiation Phase
  - Planning Phase
  - Implementation/Execution Phase
  - Closure Phase

- **Initiation Phase:**
  - The project need is identified in the initiation phase. It signals the start of a project.
- **Planning Phase:**
  - This phase is where the project solution is further explored and the steps necessary to achieve their goals.
- **Implementation Phase:**
  - The project plan is put into action. This phase is when people are carrying out the tasks of their plan.
- **Closure Phase:**
  - This phase is the final closing of all project processes and handing over deliverables to the customer.

- **Initiation Phase**
- **Example:**
- Pokhara Metropolitan decides to build a digital house-map system.
- During initiation, they only identify the need, confirm feasibility, and assign a team.
- No design or coding happens here.

- **Planning Phase**
- **Example:**
- A private bank in Kathmandu wants to introduce mobile loan applications.
- Planning includes timelines, a cost sheet, a resource plan, risk list, and communication plan.
- Show how skipping this step causes issues:
  - Budget underestimation
  - Missed deadlines
  - Confusion among developers and stakeholders

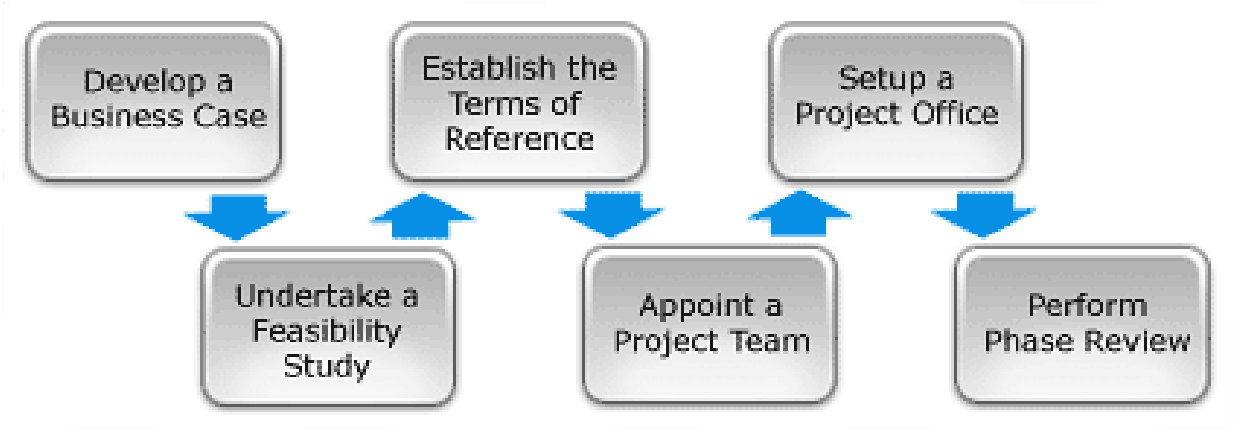
- **Execution Phase**
- **Example:**
- Developing Nepal Airlines' ticketing portal upgrade.
- Execution includes coding, testing, integrating payment gateways, security checks, and content upload.
- This phase consumes the most effort.

- **Closure Phase**
- **Example:**
- A hydropower monitoring system project in Kaski is ready for use.
- Closure includes documentation handover, final meeting, acceptance letter, and training.

Project Life Cycle

Project Initiation Phase

- The **Project Initiation Phase** is the 1st phase in the *Project Life Cycle*, as it involves starting up a new project.
- One can **start a new project by defining its objectives, scope, purpose and deliverables** to be produced.
- One will also **hire project team**, **setup the Project Office** and **review the project**.



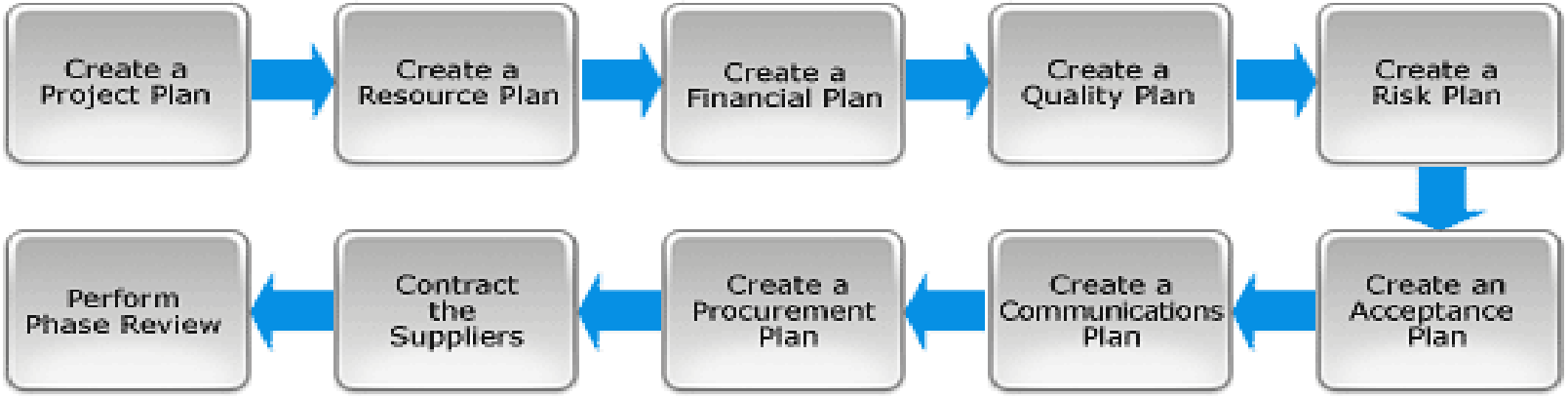
Initiation

TABLE 1.2 Initiation Phase

Phase	*Task Description*	*Output*
Initiation	✔ Broad direction for the project is provided to the project leader by the sponsor. Limits, constraints, and project priorities are defined.	Charter

Project Planning Phase

- The **Project Planning Phase** is the second phase in the *project life cycle*.
- It involves creating of a set of plans to help guide the team through the execution and closure phases of the project.
- The plans created during this phase **will help to manage time, cost, quality, change, risk** and issues.
- They will also help **manage staff and external suppliers**, to ensure that you deliver the project on time and within budget.



Planning

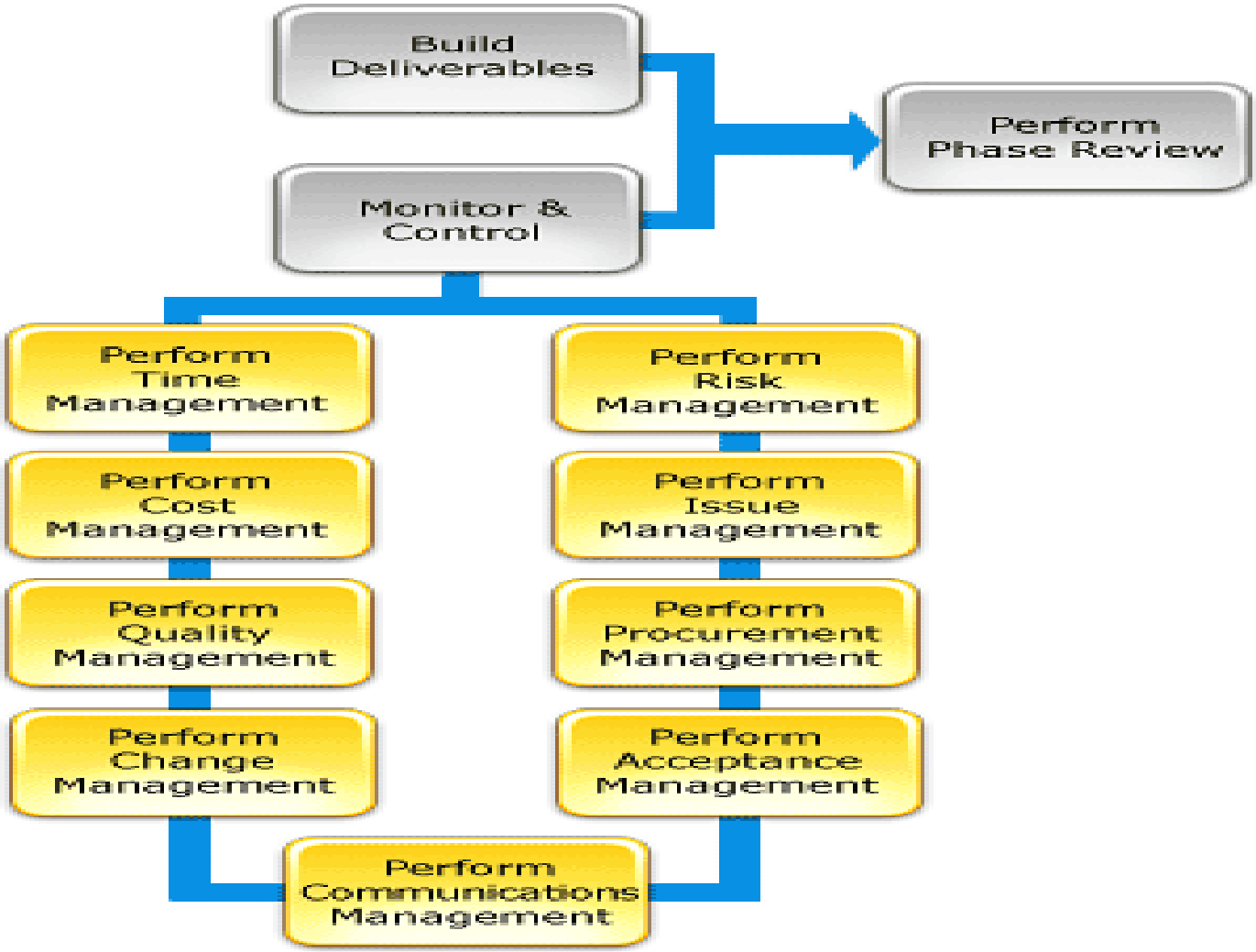
TABLE 1.3 Planning Phase

Phase	*Task Description*	*Output*
Planning	✔ Select the team members ✔ Define the scope of the project ✔ Define any risks associated with the project and develop ways to prevent them ✔ Determine the resources required to complete the project	Project Plan

Project Execution Phase

- The **Project Execution Phase** is the third phase in the *project life cycle*.
- One will **build the physical project deliverables** and **present them to the customer**.
- This is usually the **longest phase** in the project life cycle and it typically **consumes the most energy and the most resources.**

Project Execution Phase

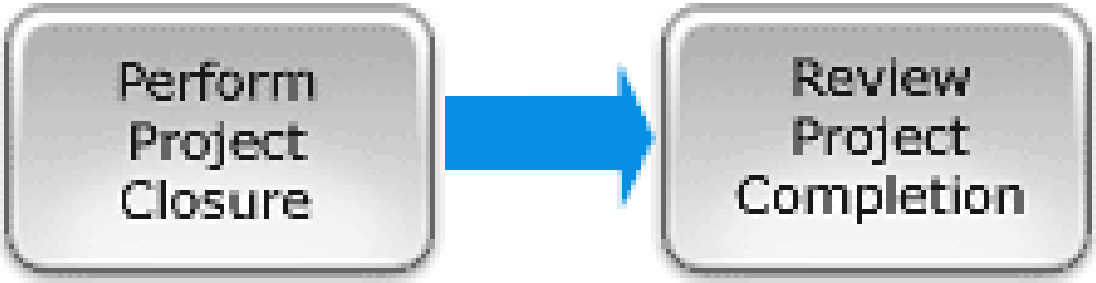


Execution

TABLE 1.4 Execution Phase		
Phase	*Task Description*	*Output*
Execution	✔ Create the deliverables	Status reports
	✔ Monitor project progress	Final deliverable
	✔ Resolve issues	
	✔ Communicate progress	
	✔ Manage changes to the plan	

Project Closure Phase

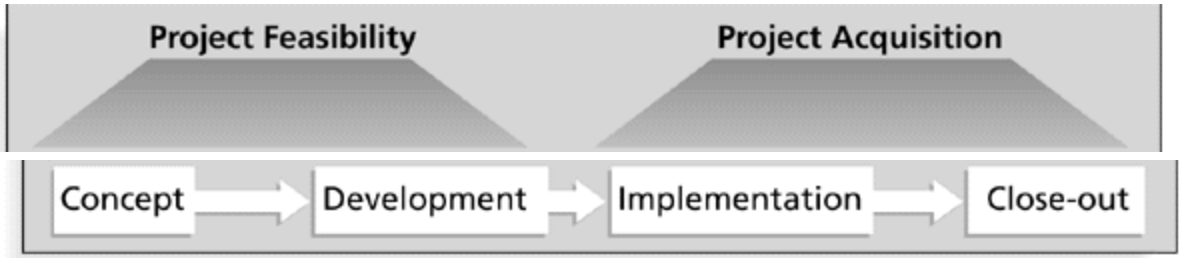
- The **Project Closure Phase** is the fourth and last phase in the *project life cycle*.
- In this phase, one will **formally close the project** and then **report its overall level of success** to the sponsor.
- Project Closure involves **handing over the deliverables to the customer**, **passing the documentation to the business**, **cancelling supplier contracts**, **releasing staff and equipment**, and **informing stakeholders of the closure of the project**.



Closure

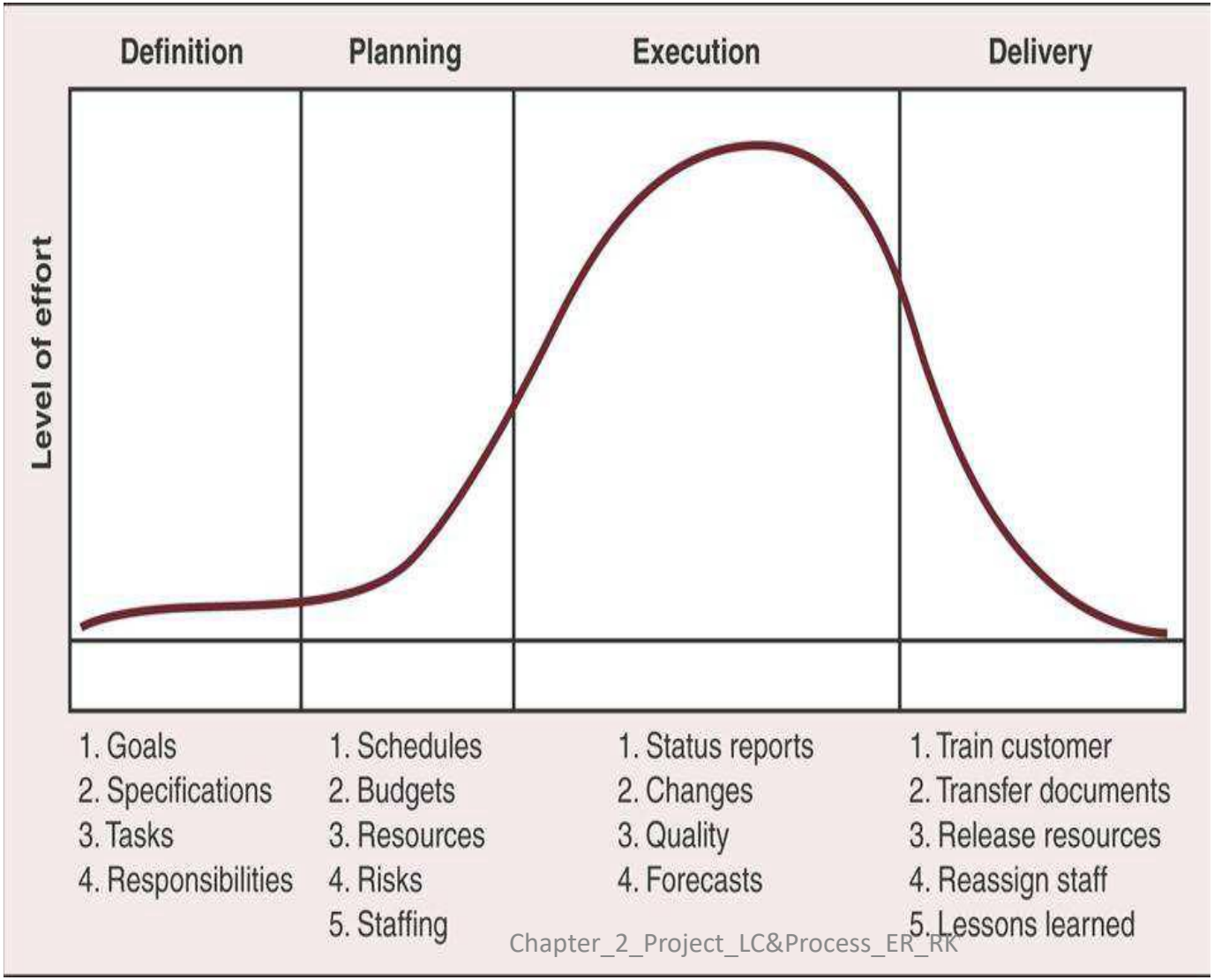
TABLE 1.5 Close-Out Phase

Phase	*Task Description*	*Output*
Close out	✔ Evaluation of customer satisfaction ✔ Assessment of lessons learned	Close-out report



Management plan	Project plan	Last work package	Completed work
Preliminary cost estimate	Budgetary cost estimate	Definitive cost estimate	Lessons learned
3-level WBS	6+-level WBS	Bulk of time should be spent here	Customer acceptance

Project Life Cycle : An Overview



Product Life Cycle

- A **product is anything that can be offered to a market to solve a problem, or to satisfy a want or need.**
- Products have a life cycle that consists of multiple stages. First the product is conceived, then developed, then introduced and managed in the market, and finally the product is retired when the need for it diminishes.
- A **project is a temporary endeavor that is undertaken to create a unique product or service. With a project, there is a clear definition of what needs to be delivered by a specified date in time.**
- It's important to note that a product can only be developed within the context of a project, and multiple projects can occur within a product's life cycle.
- The product life cycle starts with the idea of a new product in an organization that gets evaluated during the innovation phase which is followed by the product development phase.
- After the initial market introduction, the product is adapted and then sustained and finally reaches its end of life

Product Life Cycle

- The product life cycle ***starts with the idea of a new product*** in an organization that gets evaluated during the innovation phase, which is followed by the product development phase.
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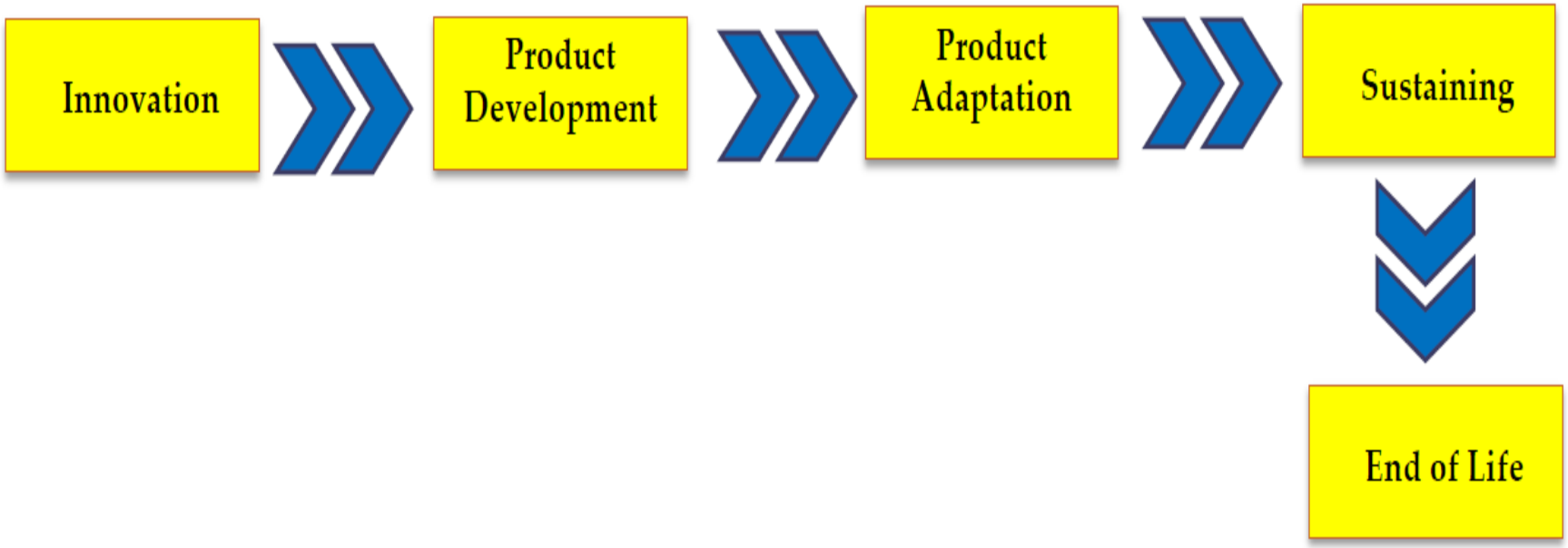
- *Phases of product life cycle*



Product Life Cycle

Review Milestone	Milestone Questions
End Of Innovation	Does a product idea exist that is ready for production? Is the available product idea good for the company and its target market?
End of Product Development	Is the implementation done correctly? Is the product functionality meeting customer needs? Is the software product ready for delivery? Is marketing ready to launch the product?
End of product Adaptation	Do existing product modifications meet customer demands?
End of Sustaining	How big is the existing customer base that is actively using this product? Is this customer base ready to transfer to a new product? Can resources supporting this product be shifted to other product developments?
End of life	Have all customers been transferred to the new product? Has the infrastructure, the product and its documentation been archived?

Stepwise Project Planning Process

Step 0: Select project

Step 1: Identify project scope and objectives

Step 2: Identify project infrastructure

Step 3: Analyze project characteristics

Step 4: Identify project products and activities

Step 5: Estimate effort for each activity

Step 6: Identify activity risks

Step 7: Allocate resources

Step 8: Review/publicize plan

Step 9: Execute plan

Step 10: Execute lower levels of planning

Project Planning

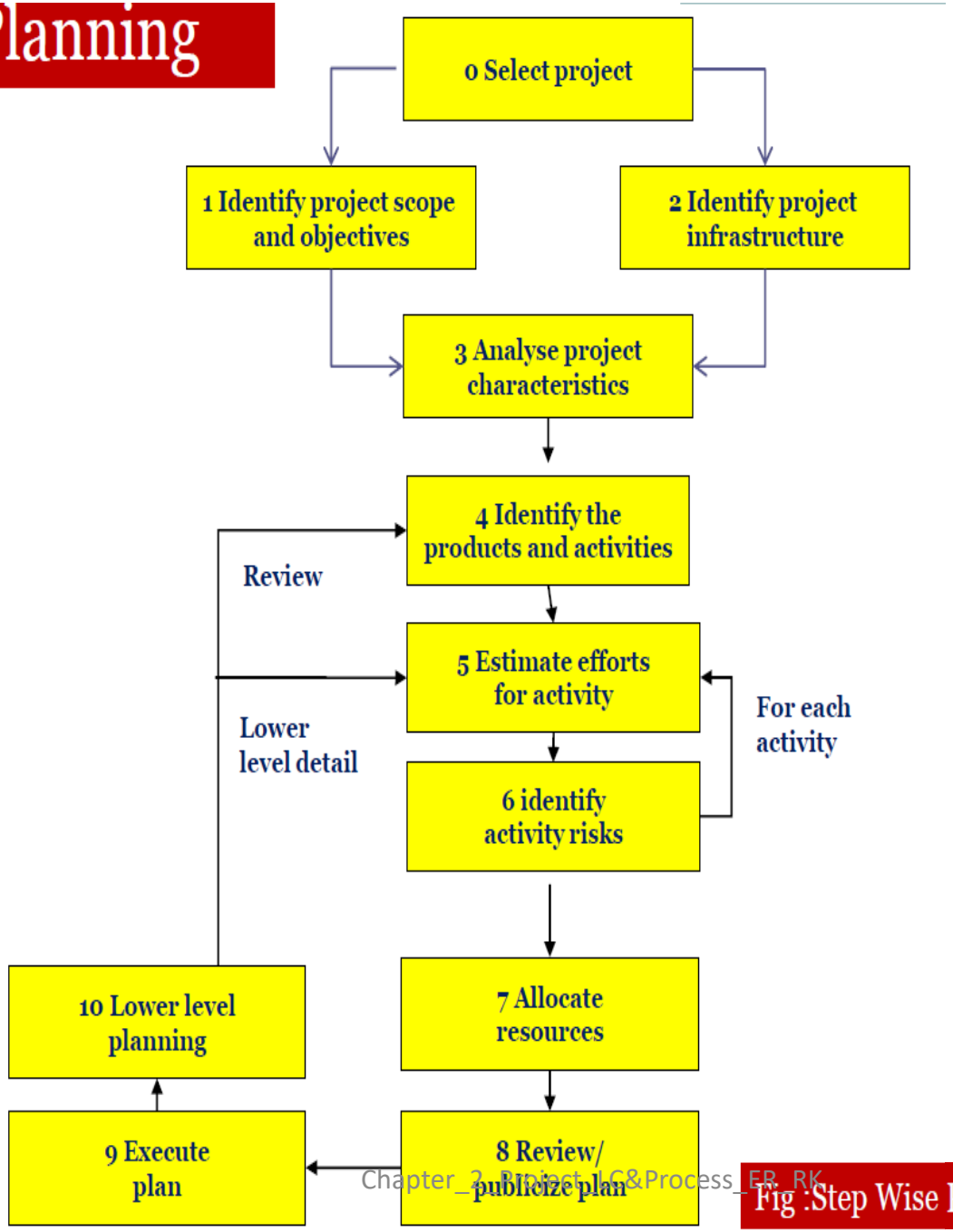


Fig :Step Wise Planning Activity

Step 0: Select Project

- This is called step 0 because in a way of project planning, it is outside the main project planning process.
- Feasibility study suggests us that the project is worthwhile or not.

Step 1: Identify project scope and objectives

- Identify objectives and measures of the effectiveness in meeting those objectives
- Establish a project authority.
  - To ensure the unity of purpose among all persons concerned.
- Identify all stakeholders in the project and their interests
- Modify objectives in the light of stakeholder analysis if necessary
- Establish methods of communication between all parties

Step 2:Identify Project Infrastructure

- **Identify relationship between the project and strategic planning**
  - To determine the order of related projects (in the organization) being carried out.
  - To establish a framework within which the system fits.
  - To ensure the hardware and software standards are followed.
- **Identify installation standards and procedures.**
  - More appropriate name: “Identify standards and procedures related to the software project.”
- **Identify project team organization**

Step 3: Analyze Project Characteristics

- Distinguish the project as either objective-driven or product driven.
- Analyze other project characteristics (including quality-based ones)
- Identify high level project risks
- Take into account user requirements concerning implementation
- Select general lifecycle approach in the light of the above.
- Review overall resource estimates
  - Up to this stage,
    - the major risks of the project are identified
    - the overall approach of the project is decided
  - So, it is a good place to re-estimate the required effort and other resources for the project

Step 4: Identify Project Products and Activities

- Identify and describe project products.
  - Identify all the products related to the project.
  - Account for the required activities.
- Document generic product flows
  - To document the relative order of the products.
- Recognize product instances
- Produce an ideal activity network
  - Activity network shows the tasks that have to be carried out as well as their sequence of execution for the creation of a product from another
- Modify the ideal to take into account need for stages and checkpoints.
  - To check compatibility of products of previous activities.

Step 5: Estimate Effort for Each Activity

- Need to estimate staff effort, time for each activity, and other resources
- Need to break a task into a series of manageable sub-tasks

Step 6: Identify Activity Risks

- Identify and quantify the risks of each activity
- Plan risk reduction and likelihood measures where appropriate.
- Adjust overall plans and estimates to take account of the risks.

Step 7: Allocate Resources (Staffing)

- Type of staff needed for each activity are identified
- Staff availabilities are identified, staff are conditionally allocated to task
- Revise plans and estimates to take into account resource constraints

Step 8: Review/publicize Plan

Review quality aspects of the project plan

- To ensure each activity is completed with a quality product

Document plans and obtain agreement

- all parties understand and agree to the commitments in the plan

Step 9: Execute Plan

- Build the physical project deliverables and present them to the customer.
- Once the project is underway, plans will need to be drawn up in greater detail for each activity as it becomes due. Detailed and lower level of planning of the later stages will need to be delayed because more information will be available nearer the start of the stage. Project planning is an iterative process. As the time approaches for the particular activities to be carried out they should be re-planned in more detail.

Stepwise Project Planning

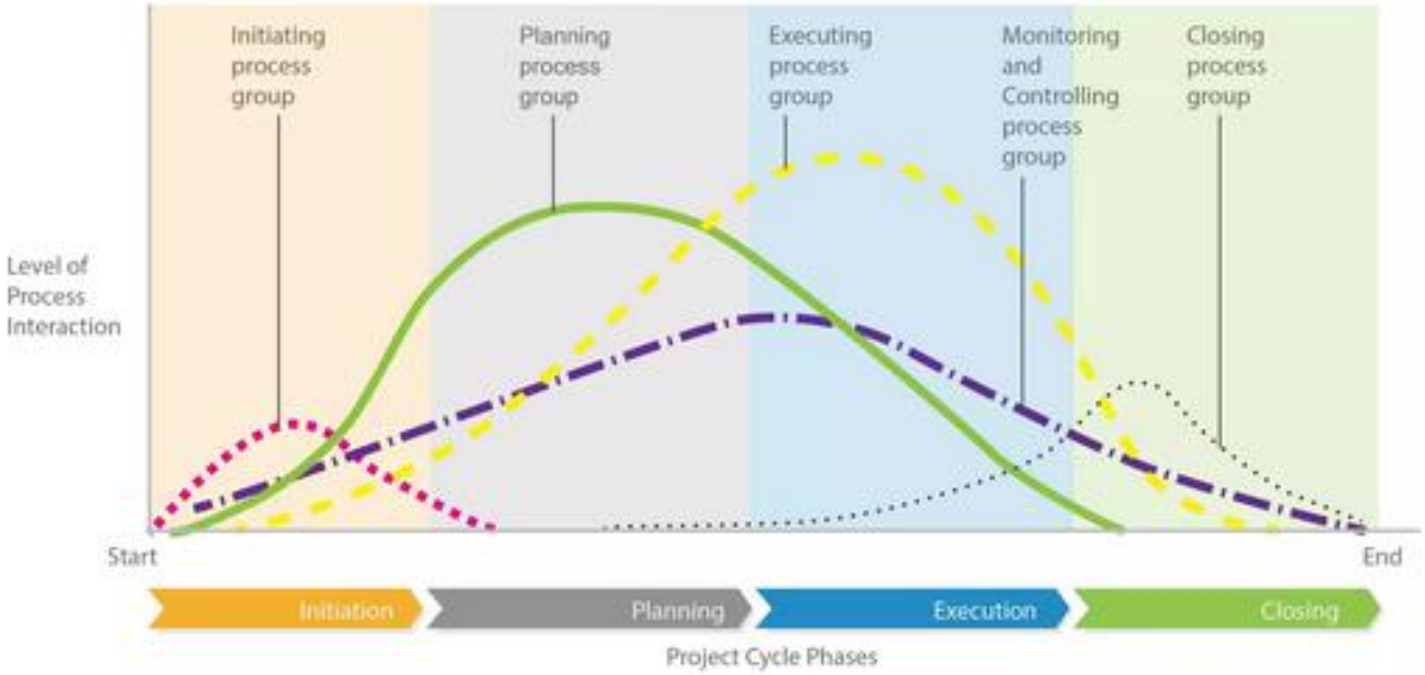
- Step Wise project planning framework
- Preparation of a software project plan
- Planning and scheduling the activities in software project management
- Various approaches towards activity plan
- Various scheduling techniques such as sequencing and CPM

Processes

- Processes are the tasks and methodologies that occur within a phase.
- Although they sound like phases, processes are not phases.
- The PMBOK (Project Management Body of Knowledge) arrange processes into groups according to when they typically occur in a project.
- However, different processes can occur any time during a project phase.

Processes

- This graphic is a representation of when these processes typically occur over the project life cycle



Initiating Process Group

- Typically occurring during the initiation phase of a project.
- These process groups define the new project or phase objectives and what is needed to achieve it.
- Involves obtaining authorization for the project or phase.
- Process group main output is the project charter and stakeholder register.

Initiation processes:

4.1 Develop Project Charter

13.1 Identify Stakeholders

Planning Process Group

- Typically occurring during the planning phase of a project.
- Planning process group does not just apply to the planning phase. Planning is an activity that continues to almost the end of a project, as projects tend to change as it progresses.
- A skilled project manager knows how much time and effort to put into each stage of planning as the project changes

Planning processes:

4.2 Develop Project Management Plan
5.1 Plan Scope Management
5.2 Collect Requirements
5.3 Define Scope
5.4 Create WBS
6.1 Plan Schedule Management
6.2 Define Activities
6.3 Sequence Activities
6.4 Estimate Activity Resources
6.5 Estimate Activity Duration
6.6 Develop Schedule
7.1 Plan Cost Management
7.2 Estimate Costs
7.3 Determine Budget
8.1 Plan Quality Management
9.1 Plan Human Resource Management
10.1 Plan Communications Management
11.1 Plan Risk Management
11.2 Identify Risks
11.3 Perform Qualitative Risk Analysis
11.4 Perform Quantitative Risk Analysis
11.5 Plan Risk Responses
12.1 Plan Procurement Management
13.2 Plan Stakeholder Management

Executing Process Group

- Typically occurring during the implementation/execution phase of a project.
- This process group involves a lot of management tasks and coordination with staff while the project plan is being executed.
- The outputs produced during this process group is the actual project deliverable.

Execution processes:

4.3 Direct and Manage Project Work
8.2 Perform Quality Assurance
9.2 Acquire Project Team
9.3 Develop Project Team
9.4 Manage Project Team
10.2 Manage Communications
12.2 Conduct Procurements
13.3 Manage Stakeholder Engagement

Closing Process Group

- Typically occurring during the closure phase of a project.
- Formal transfer of deliverable to customer
- Administrative closure and contract closure is performed to formally close out project.

Closing processes:

4.6 Close Project or Phase

12.4 Close Procurements

Monitoring and Controlling Process Group

- This process group occurs on all phases of the project life cycle.
- Ensures that the project is on track and flags issues to maintain project course direction.
- Collects performance data through constant reports from team such as status reports and team meetings.

Monitoring and Controlling processes:

4.4 Monitor and Control Project Work

4.5 Perform Integrated Change Control

5.5 Validate Scope

5.6 Control Scope

6.7 Control Schedule

7.4 Control Costs

8.3 Control Quality

10.3 Control Communications

11.6 Control Risks

12.3 Control Procurements

13.4 Control Stakeholder Engagement

From PMBOK, 5th edition

Process Group and Knowledge Area

PROCESS GROUP AND KNOWLEDGE AREA MAPPING

Knowledge Areas	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	
9. Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	

Process Group and Knowledge Area

- While the PMBOK (Project Management Body of Knowledge) and PMI (Project Management Institute) gives an outline of a typical project, no two projects are the same.
- A good project manager knows how to choose which processes are needed for their project.
- Practicality and pragmatism is the best approach to any project. (Keep it simple)

Any Queries?